

D802

Deep Learning

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A model is trained on a cluster of 8 GPUs with a batch size of 256. For one weight, $w = 2.0$, the gradient of the loss with respect to w is 0.5, and the learning rate is 0.1. What is w after one gradient descent step?

- A. 1.95
- B. 0.05
- C. 2.05
- D. 1.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

With a learning rate of 0.01, one gradient descent step moved a weight from 0.80 to 0.83. What was the gradient?

- A. -0.03
- B. -3
- C. 0.83
- D. $+3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A team trains an image model with plain SGD at a learning rate of 1.0. The images are 224×224 scans from a hospital archive. Loss swings up and down for a few hundred steps and then becomes NaN. What is the most likely fix?

- A. Resize the scans to 512×512 .
- B. Raise the learning rate so training gets past the unstable region.
- C. Add more layers to the network.
- D. Lower the learning rate, for example to 0.01.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a computation graph built in PyTorch on a laptop GPU, $y = 3x$ and $L = y^2$. During backpropagation at $x = 2$, what is dL/dx ?

- A. 6
- B. 12
- C. 15
- D. 36

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A $32 \times 32 \times 3$ color image from a 50,000-image dataset passes through a convolution layer with 16 filters of size 5×5 , stride 1, and no padding. What is the output shape?

- A. $32 \times 32 \times 16$
- B. $28 \times 28 \times 16$
- C. $27 \times 27 \times 16$
- D. $28 \times 28 \times 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A 64×64 input passes through a 3×3 convolution with stride 1. How much padding on each side keeps the output at 64×64 ?

- A. 3
- B. 0, since stride 1 always preserves size.
- C. 1
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A convolution layer outputs a $56 \times 56 \times 64$ tensor. The team, which trains with mixed precision, adds a 2×2 max-pooling layer with stride 2. What does the pooling layer produce, and how many learnable parameters does it add?

- A. $56 \times 56 \times 64$, with 0 parameters
- B. $28 \times 28 \times 64$, with 0 parameters
- C. $28 \times 28 \times 64$, with 256 parameters
- D. $28 \times 28 \times 32$, with 0 parameters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A team stacks 10 dense layers with no activation functions between them to learn a curved decision boundary. The network has 2 million parameters and is trained for 100 epochs. Why can it not fit the curve?

- A. It needs more epochs to learn the curve.
- B. Ten layers is too many; three linear layers would fit the curve.
- C. A stack of linear layers collapses into one linear function, so it can only draw straight boundaries.
- D. It has too few parameters for the task.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A deep network with sigmoid activations in every hidden layer trains very slowly, and gradients in the early layers are close to zero. Which change most directly helps?

- A. Use tanh, since it never saturates.
- B. Add more sigmoid layers.
- C. Remove all activation functions.
- D. Switch the hidden activations to ReLU.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A utility wants to predict next hour's electricity demand from the previous 48 hourly readings. Readings come from 12 substations and are stored in a time-series database. Which architecture is built for this kind of input?

- A. A recurrent network such as an LSTM
- B. A feedforward network that treats the 48 readings as unordered inputs
- C. A 2-D convolutional network designed for images
- D. Twelve separate architectures, one per substation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Training loss keeps falling, but validation loss starts rising after epoch 12. The model is trained on 4 GPUs with a batch size of 128. What should the team do?

- A. Stop at the epoch with the lowest validation loss and consider adding dropout or weight decay.
- B. Double the batch size.
- C. Train for more epochs until validation loss comes back down.
- D. Stop tracking validation loss, since training loss is improving.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A model uses dropout with a rate of 0.5 during training. What should happen to dropout when the model makes predictions in production?

- A. Raise the rate to 0.9 for stronger regularization.
- B. Keep dropping 50% of units at random.
- C. Turn it off; the full network is used, with scaling that keeps activations consistent.
- D. Permanently delete half the neurons.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A model is trained ~~on a cluster of 8 GPUs with a batch size of 256~~. For one weight, $w = 2.0$, the gradient of the loss with respect to w is 0.5, and the learning rate is 0.1. What is w after one gradient descent step?

Target word: updated weight **Decides:** $w - 0.1 \times 0.5$.

A	Correct	$2.0 - 0.05 = 1.95$.
B	Wrong stem number	The step size, not the new weight.
C	Inverted operation	Adds the step, moving uphill on the loss.
D	Dropped element	Subtracts the raw gradient and drops the learning rate.

2. Answer B Concept 1 • transfer

With a learning rate of 0.01, one gradient descent step moved a weight from 0.80 to 0.83. What was the gradient?

Target word: solve for the gradient **Decides:** $0.83 = 0.80 - 0.01 \times g$.

A	Dropped element	The weight change, not divided by the learning rate.
B	Correct	$g = (0.80 - 0.83) \div 0.01 = -3$; the weight rose because the gradient was negative.
C	Wrong stem number	The new weight.
D	Inverted operation	A positive gradient would have lowered the weight.

3. Answer D Concept 2

A team trains an image model with plain SGD at a learning rate of 1.0. ~~The images are 224×224 scans from a hospital archive.~~ Loss swings up and down for a few hundred steps and then becomes NaN. What is the most likely fix?

Target word: fix for unstable loss **Decides:** Oscillating then exploding loss at a large learning rate.

A	Camouflage	Image size does not decide the divergence.
B	Inverted operation	Bigger steps overshoot more and diverge faster.
C	Adjacent category	Capacity is not the problem; step size is.
D	Correct	Smaller steps stop overshooting the minimum.

4. Answer D Concept 3

In a computation graph ~~built in PyTorch on a laptop GPU~~, $y = 3x$ and $L = y^2$. During backpropagation at $x = 2$, what is dL/dx ?

Target word: gradient at the input **Decides:** $dL/dx = (dL/dy)(dy/dx) = 2y \times 3$.

A	Wrong stem number	The value of y , not a gradient.
B	Dropped element	Stops at dL/dy and drops the local derivative.
C	Inverted operation	Adds the derivatives ($12 + 3$) instead of multiplying.
D	Correct	$y = 6$, $dL/dy = 12$, $dy/dx = 3$; $12 \times 3 = 36$.

5. Answer B Concept 4

A $32 \times 32 \times 3$ color image ~~from a 50,000-image dataset~~ passes through a convolution layer with 16 filters of size 5×5 , stride 1, and no padding. What is the output shape?

Target word: conv output shape **Decides:** $(32 - 5 + 0) \div 1 + 1 = 28$; depth = 16 filters.

A	Adjacent category	That would need “same” padding of 2; the stem says no padding.
B	Correct	Width and height are 28; depth equals the filter count.
C	Dropped element	Forgets the +1 in the formula.
D	Confusion set	Output depth is set by the number of filters, not input channels.

6. Answer C Concept 4 · transfer

A 64×64 input passes through a 3×3 convolution with stride 1. How much padding on each side keeps the output at 64×64 ?

Target word: solve for padding **Decides:** $(64 - 3 + 2P) + 1 = 64$.

A	Wrong stem number	The kernel size.
B	Absolute language	With no padding the output shrinks to 62×62 .
C	Correct	$2P = 2$, so $P = 1$ on each side.
D	Dropped element	That is total padding; it is split across both sides.

7. Answer B Concept 5

A convolution layer outputs a $56 \times 56 \times 64$ tensor. The team, ~~which trains with mixed precision~~, adds a 2×2 max-pooling layer with stride 2. What does the pooling layer produce, and how many learnable parameters does it add?

Target word: pooling output and parameter count **Decides:** Pooling halves each spatial dimension and has no weights.

A	Dropped element	Ignores the stride that downsamples.
B	Correct	Stride 2 halves height and width; max is a fixed rule with nothing to learn.
C	Confusion set	Treats pooling like a convolution with weights.
D	Adjacent category	Pooling works per channel; it does not reduce channel count.

8. Answer C Concept 6

A team stacks 10 dense layers with no activation functions between them to learn a curved decision boundary. ~~The network has 2 million parameters and is trained for 100 epochs.~~ Why can it not fit the curve?

Target word: why the network fails **Decides:** No nonlinear activations between layers.

A	Camouflage	Training time cannot make a linear model nonlinear.
B	Adjacent category	Any number of linear layers is still linear.
C	Correct	Nonlinear activations are what let depth add expressive power.
D	Dropped element	Ignores the missing activations, which decide the limit.

9. Answer D Concept 6 · transfer

A deep network with sigmoid activations in every hidden layer trains very slowly, and gradients in the early layers are close to zero. Which change most directly helps?

Target word: reverse: from symptom to activation fix **Decides:** Saturating sigmoids cause vanishing gradients.

A	Absolute language	Tanh also saturates at large inputs.
B	Inverted operation	More saturating layers shrink gradients further.
C	Dropped element	Loses nonlinearity; the network becomes linear.
D	Correct	ReLU does not saturate for positive inputs, so gradients flow back.

10. Answer A Concept 7

A utility wants to predict next hour's electricity demand from the previous 48 hourly readings. ~~Readings come from 12 substations and are stored in a time-series database.~~ Which architecture is built for this kind of input?

Target word: architecture for sequences **Decides:** Ordered readings over time.

A	Correct	Processes inputs in order and carries state across time steps.
B	Dropped element	Drops the order that carries the signal.
C	Confusion set	Built for spatial grids, not ordered time steps.
D	Camouflage	The substation count does not decide the architecture type.

11. Answer A Concept 8

Training loss keeps falling, but validation loss starts rising after epoch 12. ~~The model is trained on 4 GPUs with a batch size of 128.~~ What should the team do?

Target word: response to diverging curves **Decides:** Validation loss rising while training loss falls means overfitting.

A	Correct	Early stopping and regularization curb overfitting.
B	Camouflage	The hardware setup does not decide the fix.
C	Inverted operation	More training deepens overfitting.
D	Dropped element	Drops the signal that reveals overfitting.

12. Answer C Concept 8 · transfer

A model uses dropout with a rate of 0.5 during training. What should happen to dropout when the model makes predictions in production?

Target word: dropout at inference **Decides:** Dropout is a training-time regularizer.

A	Inverted operation	Regularization matters during training, not prediction.
B	Dropped element	Makes predictions noisy and inconsistent.
C	Correct	Frameworks disable dropout in evaluation mode.
D	Confusion set	Confuses dropout with pruning.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	11	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Gradient descent update	Each step moves a weight against its gradient: $w_{\text{new}} = w - \text{learning rate} \times \text{gradient}$.	Adding the gradient or forgetting the learning rate.	A weight, a gradient, and a learning rate in the same stem.
2	Learning rate stability	Too large a learning rate makes loss oscillate or explode; too small makes training crawl.	Raising the learning rate when loss is unstable.	Loss that jumps, grows, or turns into NaN.
3	Backpropagation is the chain rule	The gradient at a node is the upstream gradient multiplied by the local derivative along each path.	Stopping at the output gradient or adding local derivatives.	A small computation graph and a request for a gradient at an input.
4	Convolution output size	Output width = $(W - K + 2P) \div S + 1$, and output depth = number of filters.	Forgetting the +1, assuming size is preserved, or keeping the input depth.	Input size, kernel size, stride, padding, and filter count.
5	Pooling layers	Pooling downsamples each channel by a fixed rule and has no learnable parameters; channel count is unchanged.	Counting weights for pooling or thinking it cuts channels.	"Max pooling," a window size, and a stride.
6	Activation functions	Nonlinear activations let deep networks model curves; ReLU eases vanishing gradients that sigmoid and tanh cause by saturating.	Believing stacked linear layers can model nonlinear patterns.	Layers with no activation, or early-layer gradients near zero.
7	Matching architecture to data	CNNs exploit spatial grids such as images; RNNs and LSTMs process ordered sequences, and LSTMs handle longer dependencies.	Using an image architecture for a time series, or ignoring order.	"Previous N readings," "sentence," or "sequence" vs. "image" or "pixels."
8	Regularization and early stopping	When validation loss rises while training loss falls, stop early or regularize (dropout, weight decay); dropout is turned off at inference.	Training longer, or leaving dropout on when predicting.	Diverging training and validation curves, or a dropout rate.

Course scope (public description): Deep Learning delves into the fundamental principles, underlying mathematics, and implementation details of deep learning. The curriculum is designed to provide a robust understanding of the core concepts and methodologies essential for optimizing highly parameterized models. Key topics include gradient descent, backpropagation, and the broader framework of computation graphs. The course explores the essential modules that constitute deep learning models, such as linear, convolution, and pooling layers, along with various activation functions. The course also covers common neural network architectures, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), equipping students with the skills needed to design, implement, and optimize advanced deep learning systems. Through hands-on projects and practical applications, the course prepares students to gain the expertise to tackle real-world challenges using deep learning techniques.